

A Two-Way Link Between Pseudo-Amenability and Approximate Semi-Amenability

Automated mathematics run `fa_banach_001`, lane 3

11 August 2026

Abstract

Ghahramani and Loy ask for the connection between pseudo-amenability and approximate semi-amenability. We prove that every pseudo-amenable Banach algebra is approximately semi-amenable. Conversely, an approximately semi-amenable Banach algebra with a central approximate identity is pseudo-amenable; the approximate identity need not be bounded. It follows that the two notions are equivalent for every commutative Banach algebra. The unrestricted non-commutative converse is not settled here. Status: *candidate partial result, likely valid, pending human review.*

1 Source question and definitions

The source is F. Ghahramani and R. J. Loy, *Approximate semi-amenability of Banach algebras*, arXiv:1910.03775 [1]. Its second open problem asks the question shown in Figure 1.

7. Some open problems

1. Are the Schatten classes S_p approximately semi-amenable?
2. What is the connection between approximate semi-amenability and pseudo-amenability?
3. Suppose that A is approximately semi-amenable, and has a bounded approximate identity. What can be said further? Proposition 2.4 gives a partial answer.

Figure 1: The exact source question, cropped from arXiv PDF page 24.

Write $\mathcal{A}^\#$ for the unitization of a Banach algebra $\mathcal{A}$, $\mathbf{e}$ for the adjoined identity, $\mathcal{A}^\# \widehat{\otimes} \mathcal{A}^\#$ for the projective tensor product, and $\pi(x \otimes y) = xy$. The usual module actions are

$$a \cdot (x \otimes y) = ax \otimes y, \quad (x \otimes y) \cdot a = x \otimes ya.$$

The algebra $\mathcal{A}$ is *pseudo-amenable* if it has an approximate diagonal: a net $(m_\lambda) \subset \mathcal{A} \widehat{\otimes} \mathcal{A}$ such that, for every $a \in \mathcal{A}$,

$$a \cdot m_\lambda - m_\lambda \cdot a \longrightarrow 0, \quad \pi(m_\lambda)a \longrightarrow a. \tag{1}$$

Theorems 2.6 and 2.11 and Remark 2.7 of the source give the criterion we use: $\mathcal{A}$ is approximately semi-amenable if and only if there are nets $\xi_i, \eta_i \in \mathcal{A}^\# \widehat{\otimes} \mathcal{A}^\#$ such that

$$a \cdot \xi_i - \eta_i \cdot a \longrightarrow 0 \quad (a \in \mathcal{A}), \quad \pi(\xi_i) = \pi(\eta_i) = \mathbf{e}. \quad (2)$$

The passage from approximate semi-amenable to tensors in the actual projective tensor product uses the source's equivalence with approximate semi-contractibility.

2 Proof intuition

An approximate diagonal is one tensor that is almost central. The source criterion allows two tensors: left multiplication acts on ξ_i , while right multiplication acts on η_i . Starting with one approximate diagonal, there is a direct unitization trick that manufactures this pair, which proves the unconditional forward implication.

For the reverse direction, the mixed relation in (2) implies that $\xi_i - \eta_i$ disappears after multiplication by fixed algebra elements on both sides. A central approximate identity can therefore localize ξ_i from $\mathcal{A}^\# \widehat{\otimes} \mathcal{A}^\#$ into $\mathcal{A} \widehat{\otimes} \mathcal{A}$ without creating commutator terms. The localizers may be unbounded because they are chosen first and the tensor index is chosen afterward. This order of choices is the key point.

3 The unconditional implication

Theorem 3.1. *Every pseudo-amenable Banach algebra is approximately semi-amenable.*

Proof. Let $(m_\lambda) \subset \mathcal{A} \widehat{\otimes} \mathcal{A}$ satisfy (1), and put $u_\lambda = \pi(m_\lambda)$. Applying π to the first limit in (1) gives

$$au_\lambda - u_\lambda a \longrightarrow 0.$$

Since $u_\lambda a \rightarrow a$, it follows that $au_\lambda \rightarrow a$ as well. In $\mathcal{A}^\# \widehat{\otimes} \mathcal{A}^\#$, define

$$\begin{aligned} \xi_\lambda &= \mathbf{e} \otimes \mathbf{e} - u_\lambda \otimes \mathbf{e} + m_\lambda, \\ \eta_\lambda &= \mathbf{e} \otimes \mathbf{e} - \mathbf{e} \otimes u_\lambda + m_\lambda. \end{aligned}$$

Their products are exactly $\mathbf{e}$. For $a \in \mathcal{A}$, direct expansion gives

$$\begin{aligned} a \cdot \xi_\lambda - \eta_\lambda \cdot a &= (a - au_\lambda) \otimes \mathbf{e} + \mathbf{e} \otimes (u_\lambda a - a) \\ &\quad + (a \cdot m_\lambda - m_\lambda \cdot a), \end{aligned}$$

which tends to zero. Criterion (2) completes the proof. $\square$

4 A converse with an unbounded central approximate identity

We first isolate the cancellation hidden in the two-tensor relation.

Lemma 4.1. *Suppose ξ_i, η_i obey the first limit in (2), and set $r_i(a) = a \cdot \xi_i - \eta_i \cdot a$. Then for fixed $b, c \in \mathcal{A}$,*

$$b \cdot (\xi_i - \eta_i) \cdot c = r_i(b) \cdot c - r_i(bc) + b \cdot r_i(c) \longrightarrow 0. \quad (3)$$

Proof. The identity follows by expanding its right-hand side; each of the three terms tends to zero after multiplication by the fixed elements b, c . $\square$

Theorem 4.2. *Let $\mathcal{A}$ have a central approximate identity, not assumed bounded. If $\mathcal{A}$ is approximately semi-amenable, then $\mathcal{A}$ is pseudo-amenable.*

Proof. Take ξ_i, η_i satisfying (2). We construct approximate diagonals locally. Fix a finite set $F \subset \mathcal{A}$ and $\varepsilon > 0$. Choose a central approximate-identity element $f \in \mathcal{A}$ so that fa is close to a for every $a \in F$. Then choose another central approximate-identity element $d \in \mathcal{A}$ so that $d(fa)$ is close to fa for every $a \in F$. The choices can be made so that

$$\|dfa - a\| < \varepsilon \quad (a \in F). \quad (4)$$

No common norm bound for d and f is needed.

After d and f are fixed, choose i sufficiently late and put

$$M = d \cdot \xi_i \cdot f.$$

Multiplication by d on the first tensor factor and by f on the second shows that $M \in \mathcal{A} \hat{\otimes} \mathcal{A}$, even though $\xi_i \in \mathcal{A}^\# \hat{\otimes} \mathcal{A}^\#$. Moreover,

$$\pi(M) = d\pi(\xi_i)f = df. \quad (5)$$

Since d and f are central, for $a \in F$ we have

$$\begin{aligned} a \cdot M - M \cdot a &= d \cdot (a \cdot \xi_i - \xi_i \cdot a) \cdot f \\ &= d \cdot r_i(a) \cdot f + d \cdot (\eta_i - \xi_i) \cdot af. \end{aligned} \quad (6)$$

The first term tends to zero by (2); the second tends to zero by Lemma 4.1, applied with the now-fixed elements d and af . We can therefore choose i so that $\|a \cdot M - M \cdot a\| < \varepsilon$ for every $a \in F$. Equations (4) and (5) also give $\|\pi(M)a - a\| < \varepsilon$.

Direct the triples consisting of a finite subset of $\mathcal{A}$, a positive tolerance, and the local choices above. The resulting net (M) in $\mathcal{A} \hat{\otimes} \mathcal{A}$ satisfies (1); hence $\mathcal{A}$ is pseudo-amenable. $\square$

Corollary 4.3. *For every commutative Banach algebra, approximate semi-amenable is equivalent to pseudo-amenable.*

Proof. The forward direction from pseudo-amenable is Theorem 3.1. Conversely, Theorem 2.13 of the source says that an approximately semi-amenable algebra has left and right approximate identities. In a commutative algebra, either is a central two-sided approximate identity, so Theorem 4.2 applies. $\square$

Remark 4.4. *The unboundedness allowance is substantive. The source proves that a central bounded approximate identity upgrades approximate semi-amenable to approximate amenability. Theorem 4.2 requires only an unbounded central approximate identity and reaches the weaker, natural conclusion of pseudo-amenable.*

5 Verification, upgrade attempts, and scope

Every displayed tensor identity was expanded independently. In particular, (3) is exact, M really belongs to $\mathcal{A} \hat{\otimes} \mathcal{A}$, and the two central approximate-identity elements are fixed before the tensor index; this prevents their potentially large norms from spoiling convergence.

Four focused upgrade attempts were made. First, localizing by the separate left and right approximate identities available in every approximately semi-amenable algebra creates uncontrolled commutators when the localizers are noncentral. Second, decomposing ξ_i, η_i into their four unitization components leaves asymmetric correction terms; flip symmetrization cancels them only in the commutative case. Third, the standard

finite-corner diagonals for Schatten classes have growing projective norm, so arbitrary Schatten tails do not support the rate needed to pass to the limit. Fourth, known examples separating pseudo-amenability from approximate amenability do not separate the two notions studied here, because Theorem 3.1 puts them on the approximately semi-amenable side. The unrestricted noncommutative converse and the source’s Schatten-class question therefore remain open in this packet.

A bounded novelty search on 11 August 2026 used the exact arXiv id, title, open-question wording, and combinations of “pseudo-amenable”, “approximately semi-amenable”, “central approximate identity”, and “commutative Banach algebra”. It found the source and general pseudo-amenability literature, including [2], but no direct statement of Theorems 3.1 or 4.2, or of Corollary 4.3. This was not an exhaustive literature review.

Human review should focus on the local-to-net construction in Theorem 4.2 and on whether the three statements already occur under different terminology in later Banach-algebra literature.

References

- [1] F. Ghahramani and R. J. Loy, *Approximate semi-amenability of Banach algebras*, Semigroup Forum **101** (2020), 358–384, [doi:10.1007/s00233-020-10094-6](https://doi.org/10.1007/s00233-020-10094-6); arXiv:1910.03775.
- [2] F. Ghahramani and Y. Zhang, *Pseudo-amenable and pseudo-contractible Banach algebras*, Math. Proc. Cambridge Philos. Soc. **142** (2007), 111–123, [doi:10.1017/S0305004106009649](https://doi.org/10.1017/S0305004106009649).